

KATTERJAKK, Sweden

WMO: 020200

Lat: 68.420N

Lon: 18.168E

Elev: 517

StdP: 95.27

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.4	-23.6	-28.8	0.3	-26.0	-25.9	0.4	-23.4	15.5	-3.7	14.0	-5.9	1.1	210	0.810

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	20.6	13.8	18.4	12.9	16.4	11.8	14.7	19.0	13.6	17.3	12.5	15.7	3.3	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.9	9.8	16.1	11.7	9.1	15.0	10.8	8.6	14.1	42.9	19.1	39.7	17.4	36.9	15.7	19.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	10.9	9.2	DB	-29.0	23.8	2.3	2.2	-30.7	25.4	-32.0	26.7	-33.3	27.9	-35.0	29.6	(4)
(5)			WB	-28.9	16.2	2.2	1.3	-30.5	17.1	-31.8	17.9	-33.1	18.7	-34.7	19.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.7	-10.5	-11.1	-7.7	-3.2	1.7	7.1	11.3	9.9	5.7	-0.5	-4.9	-7.6	(6)	
	DBStd	8.88	6.78	6.72	5.33	4.10	3.52	3.42	3.40	3.11	3.21	3.87	4.54	5.77	(7)	
	HDD10.0	4044	636	590	548	396	257	100	25	40	133	327	447	546	(8)	
	HDD18.3	6967	894	823	807	646	514	336	219	262	379	585	697	804	(9)	
	CDD10.0	121	0	0	0	0	1	14	65	37	4	0	0	0	(10)	
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0	(11)	
	CDH23.3	12	0	0	0	0	0	0	11	2	0	0	0	0	(12)	
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)	
(14) Wind	WSAvg	3.3	3.3	3.2	4.0	3.4	3.4	3.2	2.9	2.8	3.1	3.1	3.0	3.6	(14)	
(15) Precipitation	PrecAvg	870	72	62	67	42	44	64	88	79	96	105	67	84	(15)	
	PrecMax	1118	238	120	173	88	77	103	165	182	215	176	158	217	(16)	
	PrecMin	699	0	6	14	15	9	18	40	30	25	34	9	6	(17)	
	PrecStd	97	51	30	40	22	19	21	37	32	46	32	41	50	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.6	3.1	3.8	6.7	16.9	20.6	24.8	22.2	15.8	10.3	5.7	4.3	(19)	
		MCWB	2.0	1.9	1.4	3.3	9.8	12.5	16.2	14.6	11.1	7.2	4.4	2.4	(20)	
	2%	DB	2.0	1.5	2.2	4.9	11.0	17.5	21.6	19.6	13.7	7.4	4.1	2.7	(21)	
		MCWB	0.8	0.5	0.7	2.5	6.5	10.9	14.5	13.9	10.0	5.7	2.9	1.3	(22)	
	5%	DB	0.6	0.2	1.1	3.5	8.1	15.2	19.4	17.1	12.0	6.1	2.6	1.1	(23)	
		MCWB	-0.2	-0.6	-0.1	1.6	4.7	9.9	13.6	12.8	9.4	4.9	1.5	-0.1	(24)	
	10%	DB	-1.4	-1.7	-0.3	2.4	6.3	13.1	17.3	15.0	10.5	4.7	1.0	-0.1	(25)	
		MCWB	-2.2	-2.5	-1.4	0.9	3.7	8.9	12.6	11.8	8.6	3.6	0.2	-1.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	2.0	2.0	3.8	9.8	13.3	16.9	16.3	12.2	7.4	4.7	3.0	(27)	
		MCDB	3.1	2.7	2.9	5.9	16.2	17.8	23.1	19.9	14.3	8.9	5.3	3.8	(28)	
	2%	WB	1.1	0.7	0.9	2.8	6.9	11.9	15.4	14.3	10.7	5.9	3.1	1.4	(29)	
		MCDB	1.8	1.3	2.0	4.6	10.5	16.4	20.4	18.0	12.6	6.8	3.9	2.6	(30)	
	5%	WB	-0.3	-0.5	-0.1	1.9	5.0	10.5	14.2	13.3	9.7	4.8	1.7	0.1	(31)	
		MCDB	0.6	0.2	1.0	3.2	7.5	14.4	18.4	16.6	11.5	5.8	2.6	0.9	(32)	
	10%	WB	-2.1	-2.3	-1.4	1.0	4.0	9.2	13.1	12.1	8.7	3.6	0.2	-1.0	(33)	
		MCDB	-1.3	-1.6	-0.4	2.2	6.0	12.4	16.5	14.8	10.4	4.6	1.0	-0.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.5	6.1	6.0	5.5	6.0	7.0	6.6	5.6	4.4	5.1	5.9	(35)	
		MCDBR	6.9	6.9	6.1	6.1	7.7	9.9	11.0	10.6	8.2	5.4	5.9	6.5	(36)	
	5% WB	MCWBR	6.2	6.3	5.3	4.6	4.6	5.3	5.4	5.6	5.1	4.1	5.0	5.6	(37)	
		MCWBR	6.6	6.5	5.9	5.7	7.2	8.6	9.8	9.1	7.1	5.0	5.4	6.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.182	0.239	0.283	0.306	0.319	0.322	0.328	0.316	0.290	0.249	0.186	N/A	(40)	
		taud	1.905	2.084	2.146	2.206	2.293	2.414	2.489	2.508	2.538	2.219	1.951	N/A	(41)	
	Ebn at Noon		83	547	741	838	870	879	862	833	769	535	69	0	(42)	
		Edh at Noon	7	45	80	102	105	97	87	76	57	38	6	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.03	0.38	1.65	3.58	4.91	4.90	4.48	3.23	1.70	0.56	0.08	0.00	(44)	
	RadStd		0.00	0.06	0.17	0.21	0.36	0.50	0.45	0.29	0.16	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page